UK Ethnic Demographic Change Under Alternative Migration and Demographic Scenarios, 2022–2122

Conditional Projections for England and Wales from Census 2021 and ONS 2022-Based Assumptions

Scott Brodie Forsyth

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Abstract

This paper presents an open, reproducible population-projection system for examining how the self-identified ethnic composition of the United Kingdom could evolve under alternative demographic assumptions over a 100-year horizon. The system implements a multistate cohort-component model with separate immigration and emigration, age-specific fertility and mortality, and a five-category migrant-generation state space. The central estimand is the *conditional demographic trajectory* generated by a specified set of assumptions—not a prediction of future population composition.

The first empirical application covers England and Wales. The mid-2022 base population of 60.3 million is constructed from Census 2021 ethnicity tables (ONS RM032), disaggregated to single year of age using official mid-year population estimates, and scaled to ONS mid-2022 totals. Mortality follows ONS National Life Tables (2020–2022); fertility is calibrated from Nomis live-birth registrations (2022) and Census ethnic fertility differentials; migration volumes follow ONS 2022-based national population projection variants (zero, low, principal, and high net international migration), with age and ethnic profiles derived from Census country-of-birth tables.

Under the principal ONS migration variant, the model projects 64.9 million residents in England and Wales by 2050, with the White share in England falling from 81.0% (2022) to 66.3%; under the high-migration variant, population reaches 72.7 million with a White share of 63.5%. Under zero net migration, population falls to 52.7 million. All outputs are conditional on stated assumptions. Extension to Scotland and Northern Ireland, historical hindcasts, probabilistic uncertainty, and partnership-based fertility remain future work.

Keywords: population projection, ethnicity, migration, cohort-component model, England and Wales, demographic scenarios, Census 2021

1 Introduction

Population projections by ethnic group are analytically important and methodologically demanding. Ethnicity in the United Kingdom is measured through self-identification on the census; categories have changed across census rounds; and the demographic behaviour associated with ethnic groups—fertility, mortality, migration, partnership, and identity transmission—differs in ways that standard national projections do not capture (Haskey, 2002; Rees et al., 2011, 2017).

Prior work demonstrates both the feasibility and the sensitivity of long-run ethnic composition trajectories to assumptions about fertility convergence and migration, including Wohland et al. (2010), Coleman (2010), and the NewEthPop system (NEWETHPOP, 2013). This work extends that tradition with an open-source pipeline updated for Census 2021 base populations, explicit preregistration of scenarios in version-controlled YAML files, separate immigration and emigration modules, and parameter calibration from official aggregate tables rather than synthetic demonstration values.

The research question, stated at UK level, is:

> *How does the projected ethnic composition of the United Kingdom change between 2022 and 2122 under alternative migration and demographic scenarios?*

The answer is always conditional: under assumptions A, the model produces trajectory T_A. The system never claims that "the UK will have" a particular ethnic composition. Outputs are labelled accordingly throughout. The present implementation reports results for England and Wales; extension to Scotland and Northern Ireland follows the same architecture once harmonised census bases are integrated.

1.1 Contributions

This paper makes four contributions. First, it specifies a transparent multistate cohort-component architecture for ethnic population projection with documented accounting identities. Second, it implements reproducible ingestion of Census 2021 and ONS/Nomis aggregate tables for England and Wales, with version-controlled ethnic harmonisation.

Third, it calibrates mortality, fertility, and migration parameters from official sources, including ONS 2022-based national population projection (NPP) migration variants (ONS, 2024d). Fourth, it reports conditional scenario results to 2050 under those assumptions, illustrating how migration volume affects both population scale and ethnic composition.

1.2 Scope and remaining limitations

Geographic coverage is currently England and Wales only. Scotland (Census 2022) and Northern Ireland (Census 2021) are not yet integrated. Parental country-of-birth tables are not yet linked, so UK-born persons with foreign-born parents cannot be distinguished from those with two UK-born parents. Partnership-weighted fertility and census-estimated child ethnicity assignment remain rule-based approximations pending linkage to census household and parental-ethnicity tables. Fertility convergence by duration of residence is specified but not yet applied in the projection engine. No historical hindcast has been completed; outputs to 2122 should not be interpreted as forecasts until validation against Census 2011 and 2021 is reported.

2 Conceptual framework

2.1 Ethnicity as self-identified and changeable

Ethnicity is treated as a self-identified social category, distinct from nationality, citizenship, religion, ancestry, and country of birth (ONS, 2021, 2023). A person may report a different ethnic identity at different censuses. Country of birth is not used as a direct proxy for ethnicity; it informs migrant-generation classification only, with the relationship documented in the harmonisation metadata.

2.2 Population state space

The population at year t is represented as:

$$N_{n,e,g,s,a,t} \tag{1}$$

where n indexes UK nation, e ethnic group, g migrant generation, s sex, and a age (0–100+). Generation categories are:

1. foreign-born, arrived as adult;
2. foreign-born, arrived as child;
3. UK-born with two foreign-born parents;

4. UK-born with one foreign-born parent;

5. UK-born with two UK-born parents.

Generation is not inferred from ethnicity alone. In the current England/Wales base, generation shares within each ethnic group are assigned from Census 2021 tables RM010 (ethnic group by country of birth) and RM011 (country of birth by age).

2.3 Geographic scope

England and Wales are modelled as separate nations with zero net internal migration in the current implementation. Scotland and Northern Ireland will be added when harmonised census bases are available. Census day was 21 March 2021; the projection base year is mid-2022, with counts adjusted using ONS mid-year population estimates.

3 Projection model

3.1 Cohort-component transition

For existing cohorts, the core transition equation is:

$$N_{n,e,g,s,a+1,t+1} = \sum_{e',g'} N_{n,e',g',s,a,t} \cdot S_{n,e',g',s,a,t} \cdot Q_{e' \to e,t} + I_{n,e,g,s,a,t} - O_{n,e,g,s,a,t} + M^{int}_{n,e,g,s,a,t} \quad (2)$$

where S is survival probability, Q is the ethnic-identification transition matrix, I is immigration, O is emigration, and M^{int} is net internal movement between UK nations. Deaths D are implicit in $(1 - S)$ and appear explicitly in the accounting identity (Section 7).

3.2 Birth model

Births depend on maternal age-specific fertility rates (ASFR), not aggregate total fertility rate (TFR) alone:

$$B_{n,e,g,s,t} = \sum_{a,e_m,g_m,e_p,g_p} N^F_{n,e_m,g_m,a,t} \cdot f_{n,e_m,g_m,a,t} \cdot P(e_p, g_p \mid e_m, g_m, n, t) \cdot P(e, g \mid e_m, e_p, g_m, g_p, n, t) \cdot P(s \mid t) \quad (3)$$

where $P(s \mid t)$ is the sex ratio at birth (51.2% male, following ONS convention). Partnership affects fertility inputs through $P(e_p, g_p \mid e_m, g_m)$; a full two-sex couple-formation module is deferred. Child ethnicity probabilities $P(e, g \mid \cdot)$ follow documented rule-based specifications pending census parental-ethnicity tables.

3.3 Fertility convergence (planned)

Group-specific fertility will converge toward the national schedule:

$$f_{e,g,a,t} = f_{a,t}^{UK} + \lambda_{e,g,t}\left(f_{e,g,a}^{initial} - f_{a,t}^{UK}\right), \quad \lambda_{e,g,t} = \exp(-\kappa_{e,g}\, d) \tag{4}$$

where d is duration of residence. The engine currently applies static ASFR calibrated to 2022; convergence parameters are stored in scenario files for future implementation.

3.4 Migration

Immigration and emigration are modelled separately:

$$M_{n,e,g,s,a,t}^{net} = I_{n,e,g,s,a,t} - O_{n,e,g,s,a,t} \tag{5}$$

Scenario files specify ONS NPP migration variants. Immigration age weights follow the Census non-UK-born age structure (RM011); ethnic composition follows non-UK-born counts by ethnic group (RM010). Emigration is allocated in proportion to the existing population stock, with elevated weights at working ages. Volumes are scaled from UK totals to England and Wales using the mid-2022 population share.

4 Data sources and parameter calibration

4.1 Base population (Census 2021, mid-2022)

Census 2021 RM032 provides ethnicity × sex × age band for England (E92000001) and Wales (W92000004), retrieved via the ONS Beta API and cross-validated against Nomis table C2021RM032 (dataset NM_2132). RM032 uses five broad age bands; within each band, counts are disaggregated to single year of age using the age–sex structure from Nomis mid-year population estimates (NM_2002_1, mid-2022). Nation totals are then scaled to MYE mid-2022 (England: 57.1 million; Wales: 3.1 million; combined: 60.3 million).

Broad ethnic shares at base year (England): White 81.0%; Mixed 3.0%; Asian 9.6%; Black 4.2%; Other 2.2%. Wales: White 93.8%; Mixed 1.6%; Asian 2.9%; Black 0.9%; Other 0.9%.

4.2 Mortality

Age- and sex-specific survival probabilities $p_x = 1 - q_x$ are taken from ONS National Life Tables, United Kingdom, 2020–2022 (ONS, 2024e). A common national schedule is applied across ethnic groups and generations; group-specific mortality differentials will be incorporated only where suitable evidence exists.

4.3 Fertility

National ASFR by single year of age is computed from Nomis live births in England and Wales by mother's age (NM_203_1, 2022; total births 605,342) divided by female mid-2022 population at each age. The age profile is scaled to the ONS 2022-based principal long-term TFR of 1.45 (ONS, 2024d). Ethnic-group fertility scalars are estimated from Census 2021 RM032 as the ratio of children aged 0–4 to women aged 15–49 within each ethnic group, expressed relative to the national average. Because RM032 reports five broad age bands rather than single years of age, the 0–4 proxy uses the under-25 band with documented limitations in the harmonisation metadata.

4.4 Migration scenarios

Migration volumes follow ONS 2022-based NPP long-term assumptions from year ending mid-2028 onward (ONS, 2024c): principal net migration of 340,000 per year for the UK (immigration 858,000; emigration 518,000), with low (+120,000) and high (+525,000) variants. England and Wales receive approximately 89% of UK flows, based on the mid-2022 population share. A zero-migration counterfactual sets both immigration and emigration to zero.

Table 1 summarises official data sources and calibration status.

Table 1: Official data sources and calibration status (England and Wales)

Parameter	**Source**	**Status**
Ethnicity × age × sex	ONS RM032 / Nomis C2021RM032	Calibrated
Mid-2022 population	Nomis NM_2002_1	Calibrated
Country of birth × ethnicity	ONS RM010	Calibrated
Country of birth × age	ONS RM011	Calibrated
Live births by mother's age	Nomis NM_203_1 (2022)	Calibrated
Period life tables	ONS NLT 2020–2022	Calibrated
Migration volumes	ONS NPP 2022-based variants	Scenario-specified
Ethnicity (Scotland)	NRS Census 2022	Not integrated
Ethnicity (Northern Ireland)	NISRA DT-0035	Not integrated
Parental ethnicity / partnership	Census household tables	Planned
Historical hindcast (2001–2021)	Nomis historical census	Planned

4.5 Harmonisation

Nineteen Census 2021 England/Wales ethnic categories map to five broad groups (`white`; `mixed`; `asian`; `black`; `other`) and nineteen detailed categories. The mapping table records ONS output codes (10001–50002), Nomis API codes (1–19), comparability quality, and uncertainty flags. Nomis internal codes differ from published census output codes; both are stored to prevent retrieval errors.

5 Implementation status

Table 2 summarises model components.

Table 2: Model component implementation status

Component	Status	Notes
Cohort ageing and survival	Implemented	ONS NLT 2020–2022
Immigration / emigration (fixed flows)	Implemented	ONS NPP variants
Ethnic identity matrix Q	Implemented	Fixed identity (diagonal)
Births from maternal ASFR	Implemented	Nomis 2022 + Census ethnic scalars
Child generation from parent nativity	Partial	RM010/RM011; no parental COB
Partnership-weighted fertility	Partial	Rule-based weights
Fertility convergence	Planned	Parameters stored, not applied
Scotland / Northern Ireland	Planned	Census bases pending
Probabilistic uncertainty	Planned	$\geq$ 2,000 MC draws
Historical hindcast validation	Planned	Required before 2122 reporting

6 Scenario design

Scenarios are defined in YAML configuration files specifying migration variant (`zero`, `low`, `principal`, `high`), fertility variant, nations, and identity mode. Policy-relevant scenarios modify measurable quantities—annual immigration, emigration, TFR targets, convergence rates—not opaque labels such as “strict” or “liberal.” Deterministic outputs are conditional trajectories; they are never described as confidence intervals. A planned probabilistic extension will draw parameters from $p(\theta)$ following ONS practice (ONS, 2024d).

7 Validation strategy

7.1 Accounting validation

Every projection year verifies:

$$P_{t+1} = P_t + B_t - D_t + I_t - O_t + M_t^{int} \tag{6}$$

Additional checks: non-negative counts, valid probabilities, ethnic shares summing to one. Thirty-two automated tests cover data parsers, base-population construction, calibration routines, and projection accounting.

7.2 Historical validation (planned)

Hindcasts from Census 2001 to 2011 and 2011 to 2021/2022 will be conducted before any 100-year outputs are reported as substantive findings. Metrics include MAE, RMSE, share error, and coverage. MAPE will be reported only for groups above a minimum population threshold.

8 Conditional scenario results, 2022–2050

All results below refer to England and Wales combined unless noted. The base population is 60,278,591 (mid-2022). Projections run to 2050; the research design specifies 2122, but outputs beyond 2050 require completed hindcast validation.

Table 3 compares total population and England White share at 2050 under four migration scenarios, holding fertility at the ONS principal variant (long-term TFR 1.45) and mortality at ONS 2020–2022 life tables. Net migration volumes refer to UK totals under ONS 2022-based variant definitions from year ending mid-2028 onward.

Table 3: Conditional projections to 2050 by ONS 2022-based migration variant

Scenario	**2030 pop.**	**2040 pop.**	**2050 pop.**	**White share (Eng.), 2050**
Zero migration	59.2M	56.5M	52.7M	72.2%
Low (+120k net UK)	60.1M	58.5M	55.9M	70.0%
Principal (+340k net UK)	61.7M	63.5M	64.9M	66.3%
High (+525k net UK)	63.3M	68.6M	72.7M	63.5%

Under zero migration, natural decrease dominates: population falls to 52.7 million by 2050 despite ethnic diversification through differential fertility and cohort ageing. Under the principal ONS variant, population rises modestly to 64.9 million; the White share in

England falls from 81.0% to 76.7% by 2030 and 71.4% by 2040, reaching 66.3% in 2050—a decline of approximately 15 percentage points over 28 years driven primarily by net immigration and the ethnic composition of inflows, not by extreme fertility differentials alone. The high-migration variant amplifies both total growth (72.7 million) and ethnic diversification (White share 63.5%). Asian and Mixed groups account for the largest share gains under higher migration scenarios; Black and Other groups rise more modestly in absolute share terms but remain non-negligible components of projected diversification.

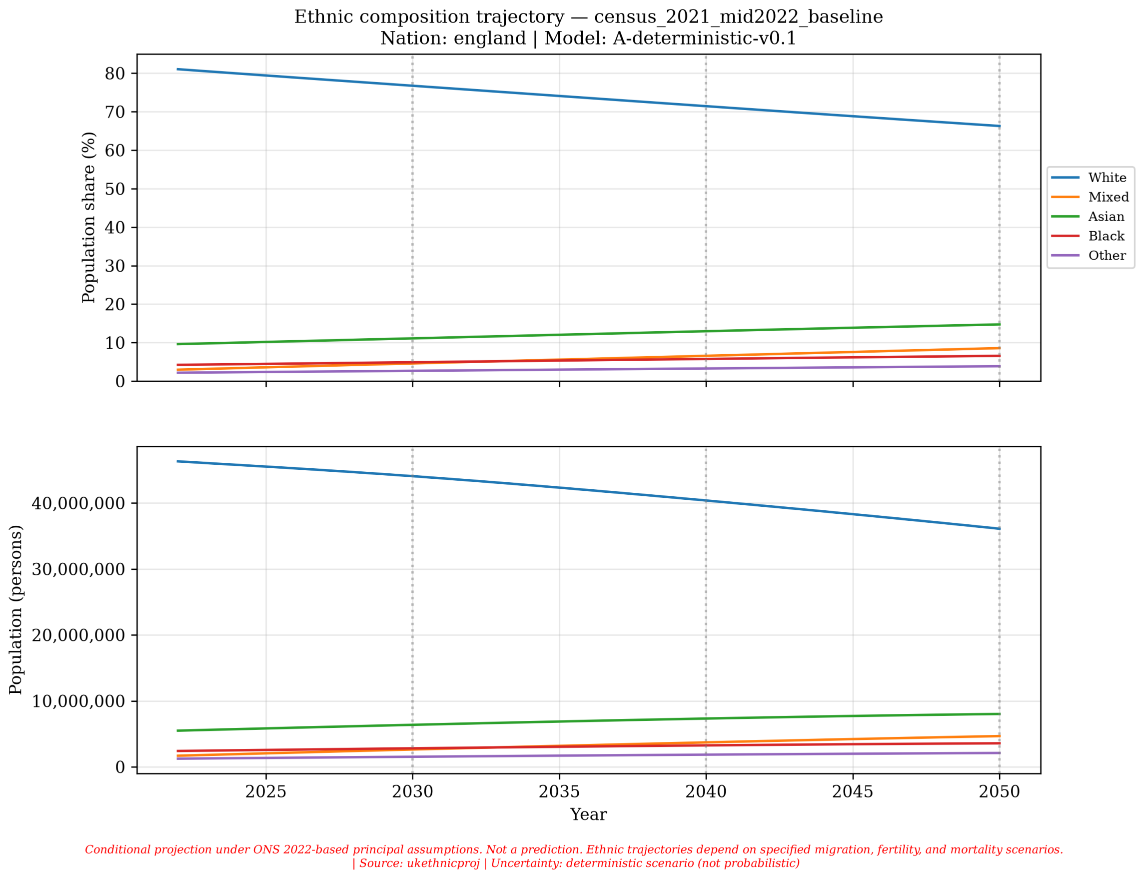

Figure 1: Ethnic population shares in England under the principal ONS 2022-based migration variant (Census 2021 base, mid-2022). Wales is included in population totals but excluded from this figure. *Conditional projection only; not a forecast.*

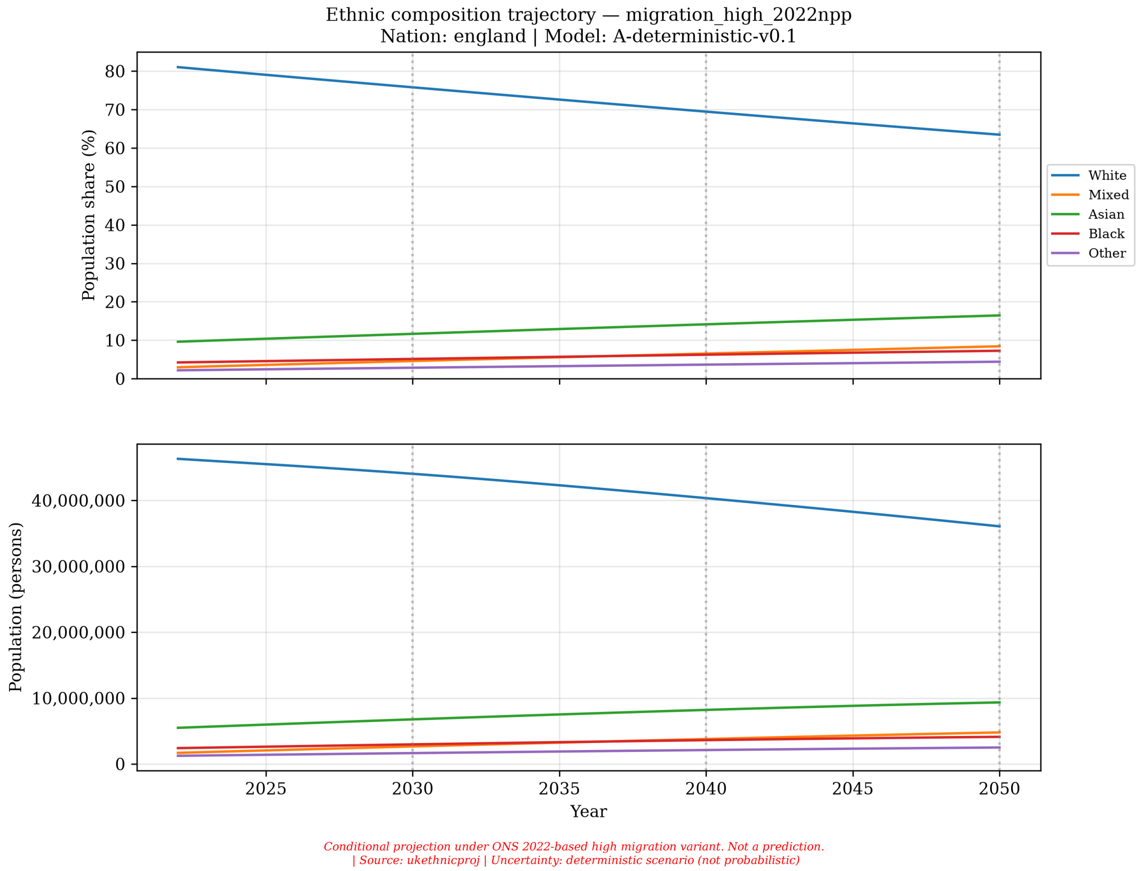

Figure 2: Ethnic population shares in England under the high ONS 2022-based migration variant. Same fertility and mortality assumptions as Figure 1; migration volume is the sole difference. *Conditional projection only; not a forecast.*

Figures 1 and 2 illustrate that migration assumptions dominate long-run ethnic composition trajectories, consistent with prior UK ethnic projection literature (Coleman, 2010; Wohland et al., 2010). The model does not attribute causal policy effects; it reports arithmetic consequences of specified demographic inputs.

9 What this study can and cannot say

Can say:

- Under ONS 2022-based principal migration assumptions, a transparent cohort-component model starting from Census 2021 projects 64.9 million England and Wales residents by 2050, with declining White share in England.
- Migration scenario choice materially affects both population scale and ethnic composition; the range across ONS variants spans 52.7–72.7 million by 2050.
- The pipeline is reproducible: all inputs are official aggregate tables with cached API responses and checksums.

Cannot yet say:

- How ethnic composition will evolve to 2122 with validated accuracy.
- Which assumptions contribute most to uncertainty (probabilistic mode not implemented).
- Whether the model would have projected Census 2011 or 2021 correctly (no hindcast).
- Results for Scotland, Northern Ireland, or the full UK total.
- Effects of specific immigration policy reforms (scenarios are demographic, not policy models).

10 Conclusion

This paper specifies and implements a reproducible ethnic population-projection system for England and Wales, calibrated from Census 2021 and ONS/Nomis official statistics. Under ONS 2022-based migration variants, conditional trajectories to 2050 show substantial sensitivity of both population scale and ethnic composition to migration assumptions—a finding consistent with prior UK ethnic projection research (Coleman, 2010; Wohland et al., 2010). The system treats ethnicity as self-identified and changeable, models immigration and emigration separately, and enforces population accounting identities at each annual step.

Three principles govern interpretation. First, all results are *conditional*: they answer specified "what if" questions rather than predicting outcomes. Second, migration assumption choice dominates long-run ethnic composition under the calibrated parameter set. Third, credibility for longer horizons requires hindcast validation and fuller UK coverage before substantive claims about 2122 are warranted.

Before such claims, historical hindcasts, Scotland and Northern Ireland integration, partnership-based fertility, and probabilistic uncertainty quantification must be completed. Until then, outputs should be read as formally specified demographic scenarios rather than forecasts of the United Kingdom's future ethnic composition.

Data and code availability

Source code, scenario configurations, harmonisation tables, and data manifests are available at
`https://github.com/sbf-developer/UK-Demographic-Change-Under-Alternative-Migration-an`

Raw API responses are cached with SHA-256 checksums under `data/raw/`. Census microdata are not used; all inputs are aggregate official tables under the Open Government Licence v3.0. Reproduction requires: `python -m ukethnicproj data fetch`, `calibrate`, `build-base-population`, and `simulate -scenario`.

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This is unfunded independent research. The author declares no conflicts of interest.

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